

# Website Traffic Analysis Report

## Project Objective

The objective of this project was to analyse website traffic data using Python to understand user behaviour, evaluate traffic source performance, identify engagement patterns, and generate actionable business insights through Exploratory Data Analysis (EDA).

## Dataset Overview

The dataset contains website traffic and user engagement metrics, including:

- Date + Hour
- Users
- Sessions
- Engaged Sessions
- Average Engagement Time per Session
- Engagement Rate
- Events per Session
- Event Count
- Traffic Source

## Tools Used

- Python
- Pandas
- Matplotlib
- Jupyter Notebook
- Power Bi

## Data Cleaning & Preprocessing

- The dataset was prepared for analysis by:
- Importing the dataset using Pandas.
- Checking data types and dataset structure.
- Handling missing values.
- Removing duplicate records.

- Converting the Date + Hour column into datetime format.
- Creating a Date column for daily trend analysis.
- Preparing the dataset for visualization and analysis.

## **Exploratory Data Analysis (EDA)**

### **1. Website Traffic Source Analysis by Users**

#### **Insight**

Organic Social is the top-performing traffic sources with 47,572 users, indicating social media is the strongest channel for attracting visitors. The company should continue investing in social media marketing. Direct, Organic Search and Referral also contribute a significant number of users. Email traffic is extremely low (only 2 users). Company should strengthen email campaigns and other underperforming channels.

### **2. Website Traffic Source Analysis by Sessions**

#### **Insight**

Organic Social is the leading traffic source with 60,627 sessions, followed by Direct and Organic Search. This shows that social media is the primary driver of website visits, while Email and Organic Video require improvement to increase user traffic.

### **3. Website Traffic Source Analysis by Engagement Rate**

#### **Insight**

Organic Video recorded the highest engagement rate (76%), indicating the highest quality of user engagement. Referral and Organic Search also performed well, while Organic Social had a comparatively lower engagement rate. The business should focus on improving engagement for high-traffic channels while maintaining the performance of high-engagement channels.

### **4. Top Traffic Sources**

#### **Insight**

Organic Social is the leading traffic source, followed by Direct, Organic Search, and Referral. These channels contribute the majority of website traffic,

indicating that organic marketing and direct visits are the primary drivers of user acquisition. Lower-performing channels such as Email, Organic Video, and Unassigned contribute very little traffic and should be reviewed for improvement

## **5. Average Session Duration by Traffic Source**

### **Insight**

Organic Video recorded the highest average session duration (180.36), showing that visitors from this channel spend the most time on the website. Referral also performs well, while Direct (45.53) and Organic Search (47.01) have the lowest session duration, indicating an opportunity to improve user engagement on these channels.

## **6. Events per Session by Traffic Sources**

### **Insight**

Website traffic fluctuated throughout the analysis period.

The highest number of sessions ( ) was recorded on, while lower traffic was observed on.

These fluctuations may be influenced by marketing campaigns, user behaviour, or website activity during different days.

## **7. Website Traffic Trend Over Time**

### **Insight**

Website sessions fluctuated across the analysis period, with noticeable peaks and dips in daily traffic.

## **Overall Findings**

- Organic Social is the top-performing traffic source, generating the highest users and sessions.
- Organic Video shows the highest user engagement, with the best engagement rate, session duration, and events per session.
- Direct, Organic Search, and Referral consistently contribute strong traffic and engagement.

- Email and Unassigned are the weakest-performing channels and need improvement.
- Website traffic fluctuates over time, indicating varying user activity and opportunities to optimize marketing efforts.

## **Overall Recommendations**

- Continue investing in Organic Social to maintain strong user acquisition.
- Increase focus on Organic Video, as it delivers the highest user engagement.
- Strengthen Email and other low-performing channels through targeted marketing campaigns.
- Improve content and landing pages for Direct and Organic Search to increase user engagement.
- Monitor website traffic regularly and optimize marketing strategies based on user behaviour and traffic trends.

## **Conclusion**

This project successfully analysed website traffic data using Python, Pandas, and Matplotlib and Power Bi to understand traffic patterns, user behaviour, and engagement across different traffic sources. Through data cleaning, exploratory data analysis (EDA), and data visualization, meaningful insights were generated to evaluate website performance and identify opportunities for improvement. The project enhanced my practical skills in data preprocessing, data analysis, visualization, and interpreting business insights. Overall, this internship project provided valuable hands-on experience in applying data analytics techniques to solve a real-world business problem and strengthened my understanding of the complete data analysis workflow.